New Wheels Project

Introduction to SQL

Problem Statement

Business Context

A lot of people in the world share a common desire: to own a vehicle. A car or an automobile is seen as an object that gives the freedom of mobility. Many now prefer pre-owned vehicles because they come at an affordable cost, but at the same time, they are also concerned about whether the after-sales service provided by the resale vendors is as good as the care you may get from the actual manufacturers.

New-Wheels, a vehicle resale company, has launched an app with an end-to-end service from listing the vehicle on the platform to shipping it to the customer's location. This app also captures the overall after-sales feedback given by the customer.

Objective

New-Wheels sales have been dipping steadily in the past year, and due to the critical customer feedback and ratings online, there has been a drop in new customers every quarter, which is concerning to the business. The CEO of the company now wants a quarterly report with all the key metrics sent to him so he can assess the health of the business and make the necessary decisions.

As a data analyst, you see that there is an array of questions that are being asked at the leadership level that need to be answered using data. Import the dump file that contains various tables that are present in the database. Use the data to answer the questions posed and create a quarterly business report for the CEO.

Business Questions

Question 1: Find the total number of customers who have placed orders. What is the distribution of the customers across states?

Solution Query 1:

```
SELECT COUNT(DISTINCT customer_id) as total_customers
FROM order_t;
```

Output:

total_customers
994

Solution Query 2:

```
SELECT ct.state, COUNT(DISTINCT ct.customer_id) AS total_customers
FROM customer_t AS ct
INNER JOIN order_t AS ot ON ct.customer_id = ot.customer_id
GROUP BY ct.state
ORDER BY total_customers DESC;
```

Output:

state	total_customers
California	97
Texas	97
Florida	86
New York	69
District of ...	35
Colorado	33
Ohio	33
Alabama	29
Washington	28
Arizona	26

Hawaii	6
Delaware	6
Arkansas	6
New Mexico	5
Montana	3
New Ham...	3
North Dak...	2
Mississippi	2
Maine	1
Vermont	1
Wyoming	1

Observations and Insights:

- The states with the highest number of unique customers placing orders are: **California** (97), **Texas** (97), Florida (86), New York (69) and so on.
- Several states, such as Maine (1), Vermont (1), and Wyoming (1), have single-digit customer participation, indicating low reach or customer interest.

Question 2: Which are the top 5 vehicle makers preferred by the customers?

Solution Query:

```
SELECT pt.vehicle_maker, COUNT(DISTINCT ot.customer_id) AS total_customers
FROM order_t AS ot
INNER JOIN product_t AS pt ON ot.product_id = pt.product_id
GROUP BY pt.vehicle_maker
ORDER BY total_customers DESC
LIMIT 5;
```

Output:

vehicle_maker	total_customers
Chevrolet	83
Ford	63
Toyota	52
Dodge	50
Pontiac	50

Observations and Insights:

- The top 5 vehicle makers based on unique customers who placed orders are: **Chevrolet** (83 customers), **Ford** (63 customers), **Toyota** (52 customers), **Dodge** (50 customers) and **Pontiac** (50 customers).
- **Chevrolet** leads by a notable margin, indicating strong brand preference, followed by **Ford**, **Toyota** & **Dodge & Pontiac**.

Question 3: Which is the most preferred vehicle maker in each state?

Solution Query:

```
WITH vehicle_preference AS (
  SELECT
    ct.state,
    pt.vehicle_maker,
    COUNT(DISTINCT ot.customer_id) AS customer_count,
    RANK() OVER (PARTITION BY ct.state ORDER BY COUNT(DISTINCT ot.customer_id) DESC) AS
   rnk
  FROM order_t ot
            INNER JOIN customer_t ct ON ot.customer_id = ct.customer_id
            INNER JOIN product_t pt ON ot.product_id = pt.product_id
  GROUP BY ct.state, pt.vehicle_maker)

SELECT state, vehicle_maker, customer_count
FROM vehicle_preference
WHERE rnk = 1
ORDER BY customer_count desc;
```

Output:

state	vehicle_maker	customer_count
Texas	Chevrolet	9
Florida	Toyota	7
California	Audi	6
California	Chevrolet	6
California	Dodge	6
California	Ford	6
California	Nissan	6
Ohio	Chevrolet	6
Alabama	Dodge	5

Vermont	Mazda	1
Wisconsin	Acura	1
Wisconsin	Cadillac	1
Wisconsin	Chevrolet	1
Wisconsin	Dodge	1
Wisconsin	Honda	1
Wisconsin	Mazda	1
Wisconsin	Nissan	1
Wisconsin	Pontiac	1
Wyoming	Buick	1

Observations and Insights:

- **California** shows equal preference for multiple brands including **Chevrolet, Dodge, Ford, Audi, and Nissan** (6 orders each).
- **Texas** has a strong preference for **Chevrolet** (9 orders).
- **Florida** prefers **Toyota** (7 orders).

Question 4: Find the overall average rating given by the customers. What is the average rating in each quarter?

Consider the following mapping for ratings: “Very Bad”: 1, “Bad”: 2, “Okay”: 3, “Good”: 4, “Very Good”: 5

Solution Query 1:

```
SELECT
    AVG(
        CASE
            WHEN ot.customer_feedback = 'Very Bad' THEN 1
            WHEN ot.customer_feedback = 'Bad' THEN 2
            WHEN ot.customer_feedback = 'Okay' THEN 3
            WHEN ot.customer_feedback = 'Good' THEN 4
            WHEN ot.customer_feedback = 'Very Good' THEN 5
        END
    ) AS average_rating
FROM order_t AS ot;
```

Output:

average_rating
3.1350

Solution Query 2:

```
SELECT
    quarter_number,
    ROUND(
        AVG(
            CASE
                WHEN customer_feedback = 'Very Bad' THEN 1
                WHEN customer_feedback = 'Bad' THEN 2
                WHEN customer_feedback = 'Okay' THEN 3
                WHEN customer_feedback = 'Good' THEN 4
                WHEN customer_feedback = 'Very Good' THEN 5
            END
        ), 4
    ) AS avg_rating
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number;
```

Output:

quarter_number	avg_rating
1	3.5548
2	3.3550
3	2.9563
4	2.3970

Observations and Insights:

- Overall Average Rating: **3.1350 out of 5**(This falls between "Okay" and "Good").
- There's a **clear downward trend** in customer satisfaction from Q1 to Q4.
- Rating dropped from **3.55 to 2.39**, which is a ~**33% decrease** in perceived service quality.
- **Q1** had the **highest rating** where as **Q4** has the **lowest rating** (2.3970).

Question 5: Find the percentage distribution of feedback from the customers. Are customers getting more dissatisfied over time?

Solution Query:

```
SELECT
    ot.quarter_number,
    ROUND(COUNT(CASE WHEN ot.customer_feedback = 'Very Bad' THEN 1 END) * 100
    COUNT(*), 2) AS very_bad_percentage,
    ROUND(COUNT(CASE WHEN ot.customer_feedback = 'Bad' THEN 1 END) * 100 / COUNT(*), 2)
    AS bad_percentage,
    ROUND(COUNT(CASE WHEN ot.customer_feedback = 'Okay' THEN 1 END) * 100 / COUNT(*),
    2) AS okay_percentage,
    ROUND(COUNT(CASE WHEN ot.customer_feedback = 'Good' THEN 1 END) * 100 / COUNT(*),
    2) AS good_percentage,
    ROUND(COUNT(CASE WHEN ot.customer_feedback = 'Very Good' THEN 1 END) * 100 /
    COUNT(*), 2) AS very_good_percentage

FROM order_t AS ot
GROUP BY ot.quarter_number
ORDER BY ot.quarter_number;
```

Output:

quarter_number	very_bad_percentage	bad_percentage	okay_percentage	good_percentage	very_good_percentage
1	10.97	11.29	19.03	28.71	30.00
2	14.89	14.12	20.23	22.14	28.63
3	17.90	22.71	21.83	20.96	16.59
4	30.65	29.15	20.10	10.05	10.05

Observations and Insights:

- **Negative feedback** ("Very Bad" + "Bad") showed a sharp increase, rising from **22.26% in Q1** to **59.80% in Q4**, indicating a significant decline in customer satisfaction over time.
- **Positive feedback** ("Good" + "Very Good") dropped substantially, falling from **58.71% in Q1** to only **20.10% in Q4**, suggesting that fewer customers were pleased with their experience in later quarters.
- **Neutral feedback** ("Okay") stayed relatively **stable** across all quarters (~20%).

Question 6: What is the trend of the number of orders by quarter?

Solution Query:

```
SELECT
  ot.quarter_number,
  COUNT(ot.order_id) AS total_orders
FROM order_t ot
GROUP BY ot.quarter_number
ORDER BY ot.quarter_number;
```

Output:

quarter_number	total_orders
1	310
2	262
3	229
4	199

Observations and Insights:

- There's a **consistent decline** in the number of orders each quarter.
- From **Q1 to Q4**, the number of orders **dropped by 35.8% ((**310 - 199**)/**310 * 100**)** overall.

Question 7: Calculate the net revenue generated by the company. What is the quarter-over-quarter % change in net revenue?

Solution Query 1:

```
SELECT
        SUM((vehicle_price * quantity) * (1 - discount)) AS total_revenue
FROM order_t;
```

Output:

total_revenue
48610993.7813

Solution Query 2:

```
WITH revenue_per_quarter AS (
   SELECT
      ot.quarter_number,
      SUM((ot.vehicle_price * ot.quantity) * (1 - ot.discount)) AS total_revenue
   FROM order_t ot
   GROUP BY ot.quarter_number
)


SELECT
   quarter_number,
   total_revenue,
   LAG(total_revenue) OVER (ORDER BY quarter_number) AS previous_revenue,
   ROUND(
      (total_revenue - LAG(total_revenue) OVER (ORDER BY quarter_number))
      / NULLIF(LAG(total_revenue) OVER (ORDER BY quarter_number), 0) * 100, 2
   ) AS qoq_percentage_change

FROM revenue_per_quarter
ORDER BY quarter_number;
```

Output:

	quarter_number	total_revenue	previous_revenue	qoq_percentage_change
▶	1	18032549.8996	NULL	NULL
	2	13122995.7562	18032549.8996	-27.23
	3	8882298.8449	13122995.7562	-32.32
	4	8573149.2806	8882298.8449	-3.48

Observations and Insights:

- Net revenue is **declining steadily across quarters**.
- **Q1 had the highest revenue**, and by Q4 it dropped by **over 52%** from Q1.
- **Q4** revenue slightly stabilized with a **smaller drop** (-3.48%).
- The biggest dip occurred **from Q2 to Q3** (-32.32%), suggesting a sharp revenue fall mid-year.
- Total Revenue is: 48610993.7813

Question 8: What is the trend of net revenue and orders by quarters?

Solution Query:

```
SELECT
    quarter_number,
    COUNT(order_id) AS total_orders,
    SUM((vehicle_price * quantity) * (1 - discount)) AS total_revenue
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number;
```

Output:

quarter_number	total_orders	total_revenue
1	310	18032549.8996
2	262	13122995.7562
3	229	8882298.8449
4	199	8573149.2806

Observations and Insights:

- Both **order volume and net revenue consistently declined** from Q1 to Q4.
- The steepest revenue drop occurred between **Q1 and Q3**, aligning with the **reduction in orders**.
- Orders dropped by ~**36%** (from 310 in Q1 to 199 in Q4).
- Net revenue dropped by ~**52%** (from 18M to 8.5M).

Question 9: What is the average discount offered for different types of credit cards?

Solution Query:

```
SELECT
   ct.credit_card_type,
   ROUND(AVG(ot.discount) * 100, 2) AS avg_discount_percentage
FROM order_t ot
         INNER JOIN customer_t ct USING (customer_id)
GROUP BY ct.credit_card_type;
```

Output:

credit_card_type	avg_discount_percentage
laser	64.38
mastercard	62.95
maestro	62.42
visa-electron	62.35
china-unionpay	62.22
instapayment	62.06
americanexpress	61.63
diners-club-us-ca	61.46
diners-club-carte-bl...	61.45
switch	61.02
bankcard	60.95
jcb	60.74
visa	60.08
diners-club-enroute	59.98
solo	58.50
diners-club-internat...	58.40

Observations and Insights:

- Discounts vary between **58.4% and 64.38%**, indicating fairly consistent discounting strategies.
- **Laser card users** received the **highest average discount** (64.38%).
- **Diners Club International** and **Solo** users received the **least average discounts**, under 59%.

Question 10: What is the average time taken to ship the placed orders for each quarter?

Solution Query:

```
SELECT
   quarter_number,
   ROUND(AVG(DATEDIFF(ship_date, order_date)), 2) AS avg_shipping_days
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number;
```

Output:

quarter_number	avg_shipping_days
1	57.17
2	71.11
3	117.76
4	174.10

Observations and Insights:

- **Consistent increase** in shipping duration every quarter.
- From **Q1 to Q4**, the average shipping time **tripled** (57 → 174 days).
- **Sharpest increase** occurred between Q2 and Q3 (71 → 117 days).

Business Metrics Overview

Total Revenue	Total Orders	Total Customers	Average Rating
48,610,993.7813 (Sum of revenue from Q1-Q4 **or** sum of revenue from the order table)	**1000** (Sum of orders from Q1-Q4 **or** Count of all order_id in the Order table)	**994** (Count of unique customer_id from the Customer table)	**3.14 / 5** Based on mapping feedback: "Very Bad" = 1 to "Very Good" = 5
Last Quarter Revenue	**Last quarter Orders**	**Average Days to Ship**	**% Good Feedback**
8,573,149.28 (Revenue from Q4)	**199** (Orders placed in Q4)	**97.96 days** (Calculated as the average of DATEDIFF(ship_date, order_date))	**Around 44.10%** Sum of ***Good Feedback*** / **Total Feedback** * ***Good Feedback*** = Good + Very Good (Very Bad- 175, Bad- 182, Okay- 202, Good- 215, Very Good- 226)

Business Recommendations

- **Operational Improvements**
  - Streamline logistics & shipping: Reduce average shipping time to below 60 days. Introduce delivery benchmarks and partner with efficient logistics providers.
  - Implement a real-time tracking system for customer transparency.
- **Marketing & Customer Retention**
  - Focus on customer re-engagement in high-performing states (CA, TX, FL).
  - Launch targeted regional campaigns for popular brands in their top-performing states.
  - Improve after-sales service and encourage feedback with incentives.
- **Revenue Optimization**
  - Reevaluate discount strategy: Avoid over-reliance on high-discounting. Introduce loyalty points or bundled offers instead of flat % cuts.
  - Introduce premium tiers or upgrades for frequent buyers.
- **Customer Satisfaction**
  - Set up customer support escalation channels to address issues faster.
  - Focus on increasing the 4-5 star rating share through better experience.
  - Post-delivery follow-up surveys to understand dissatisfaction causes.
- **Performance Monitoring**
  - Track QoQ metrics closely: Set alerts for sharp changes in shipping time, ratings, or orders.
  - Build dashboards to regularly analyze customer location, preferences, delivery times.